Skin regeneration underlies cancer resistance in Acomys

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Title: Skin regeneration underlies cancer resistance in *Acomys*

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Abstract: Cancer and regeneration share common characteristics, including cell dedifferentiation, proliferation, migration, and immune cell activation[1–3]. These processes proceed, however, during tumorigenesis without the strict control exhibited during regeneration[4]. Most common in animals without an advanced adaptive immune system, regeneration is not widespread in mammals[5]. The Spiny Mouse (*Acomys* spp.) is a notable exception, possessing the unique ability to regenerate large wounds in the skin, ear, and other complex organs completely and without the presence of scar tissue[6–8]. Here, we employ *Acomys* to test the hypothesis that enhanced mammalian regeneration significantly alters cancer incidence and/or initiation compared to the non-regenerative laboratory mouse (*Mus musculus*). We found that *Acomys* has drastically reduced tumor incidence and increased survival following chemical carcinogenesis compared to *Mus,* despite a prolonged exposure period. *Acomys* undergoes a regenerative response including widespread cell death and increased epithelial cell proliferation in response to chemical carcinogenesis. During this process, we identify a subset of treatment-dependent Tenascin C (*Tnc*)-expressing regenerative fibroblasts and Tumor necrosis factor (*Tnf*) expressing T cells as the likely drivers of regeneration. Finally, we found that Erk phosphorylation, a key molecular mechanism of tissue regeneration in *Acomys*, is a significant mediator of epidermal cell death following carcinogenesis. These findings establish a new perspective on cancer initiation and the mechanistic link between tissue regeneration and intrinsic cancer resistance.

Main text: Epimorphic regeneration is the process through which multi-lineage tissues functionally regenerate following an injury, either with or without the formation of an undifferentiated cell blastema [9]. This type of regeneration is found in invertebrate planarian flatworms, as well as reptiles and amphibians, which are capable of limb regeneration [10–12]. Except for fingertip digits, adult mammals are generally incapable of complex tissue regeneration. Only a select few mammals, such as spiny mice and reindeer, possess the ability for more advanced tissue regeneration [6,13,14]. Mammalian wound healing has been linked to the formation of tumors from their cancer cells of origin, and the similarities between non-regenerative wound healing and tumor hallmarks have been explored in depth [15,16]. However, epimorphic regeneration was postulated as a mechanism of tumor resistance as early as 1935 [17]. This has been the basis for studies in urodeles, which have relatively low natural tumor occurrence and can eliminate malignant cells during the regeneration of amputated limbs [18,19]. Despite this, mechanisms of cancer resistance stemming from epimorphic regeneration remain poorly understood, especially in mammalian systems where this type of regeneration is limited [1].

Cancer incidence is highly variable within the animal kingdom, ranging from zero to greater than 50% [20,21]. In humans, a mammal with little to no epimorphic regeneration capability, cancer remains the second leading cause of death both in the United States and worldwide [22]. The Spiny Mouse, *Acomys* spp., possesses remarkable epimorphic regeneration and wound-healing abilities. The incredible non-scarring regenerative propensity of *Acomys* has been described in many organs, including the heart, skeletal muscle, kidney, spinal cord, and skin – a phenomenon not described elsewhere in mammalian phylogeny [6–8,23–25]. In *Acomys* and other phyla-spanning regenerative species, several tumor suppressor genes and pathways have been attributed to the tightly controlled cell proliferation and patterning required for epimorphic regeneration, and their expression has posed difficulties in dedifferentiating *Acomys* cells into induced pluripotent cells *in vitro* [2,26–29]. We thus hypothesized that the regenerative capacity of *Acomys* may contribute to a tumor-resistant phenotype. There are no documented cases of tumor incidence in captive or wild *Acomys;* however, experimental evidence confirming mechanistic *in vivo* tumor resistance in *Acomys* has remained undescribed. Here, we employed an *in vivo* chemical carcinogenesis model to define and identify differences in tumor incidence and the initial response to carcinogenic events between *Acomys* and the non-regenerative laboratory mouse *Mus musculus*.

Reduced cutaneous tumor incidence in *Acomys*

We hypothesized that the induction of skin tumors would be low to absent in *Acomys*, which is capable of blastema-independent cutaneous epimorphic regeneration, compared to *Mus*, where tumors form at a relatively high incidence. To test the susceptibility of *Acomys* to cutaneous carcinogenesis, we utilized a well-established two-stage chemical carcinogenesis method [30]. The dorsal skin of adult animals was shaved and treated with a single application of dimethylbenz[a]anthracene (DMBA) for a tumor-initiating event, followed by tumor promotion by 12-O-tetradecanoylphorbol-13-acetate (TPA). Tumor promotion with topical TPA application began one week after DMBA and was performed twice weekly throughout the remainder of the study period (Figure 1A). Along with *Acomys*, DMBA/TPA treatment was performed on two mouse strains, FVB/N and C57Bl/6J, each with differing susceptibilities to cutaneous tumor development (highly susceptible and moderately susceptible, respectively). As expected, both FVB/N and C57Bl/6J mice were prone to tumor formation and developed lesions after six and seven weeks of treatment (Figures 1B-C). The majority reached a humane endpoint at 11 weeks (FVB/N) and 23 weeks (C57Bl/6J). Strikingly, *Acomys* did not develop comparable tumors during one year of TPA treatment. In *Acomys*, very small areas of epithelial hyperplasia with little to no progression began developing only well after 17 weeks of treatment, and 50% of animals remained completely lesion-free through up to one year of TPA administration (Figure 1B-C, Extended Data Figure 1), indicating a remarkable degree of tumor resistance.

The *Acomys* lesions that developed were infrequent, small, and slow-growing, and no animal reached the established clinically humane endpoint during a year-long treatment period (Figure

1D). For these studies, we used a highly conservative measure of tumor development by including any animal with a macroscopically visible skin hyperplasia. Despite this, the overall *Acomys* tumor burden was substantially lower, with a mean of 0.75 tumors per animal, compared to both FVB/N and C57Bl/6J, with means of 23.8 and 15 tumors per animal, respectively (Figure 1E). Together, the delayed onset of tumor initiation, increased survivability, and dramatically reduced tumor burden indicate that *Acomys* are markedly resistant to DMBA/TPA-mediated skin carcinogenesis.

Similar DMBA effects on DNA damage and mutation in *Acomys* and *Mus*

Given the substantial reduction in tumor initiation and progression in *Acomys*, we investigated the initial response to DMBA to determine whether sufficient chemical penetrance for mutagenesis had occurred. To assess DMBA-mediated DNA damage in the cutaneous epithelium, we used γH2AX staining to quantify the frequency of cells undergoing DNA repair. At 24 hours post-DMBA application (Figure 2A), both C57Bl/6J and *Acomys* showed a 10-fold increase in epidermal γH2AX staining relative to untreated animals, indicating similar rates of DMBA-induced DNA damage induction (Figure 2B-C). We next examined the continued incidence of γH2AX staining seven days post-DMBA application to determine whether double-strand breaks were present when tumor promotion with TPA treatment began. At this timepoint, the number of damaged cells in both *Acomys* and *Mus* was elevated compared to untreated controls (Figure 2B-C).

Cutaneous tumor initiation occurring from DMBA reliably results from an activating mutation in *Hras*, defined by a nucleotide transversion at the second location of codon 61[31,32]. We performed Sanger Sequencing of *Acomys* in endpoint tumor tissue to identify the prevalence of this mutation. Surprisingly, no *Hras* mutation was found after one year of TPA promotion (Extended Data Figure 1D), despite identifying DNA damage foci in the early response to DMBA (Figure 2B-C). Given that the presence of DNA damage appeared quite similar between *Acomys* and *Mus,* but no mutation burden was identified at endpoint analysis, we hypothesized that *Acomys* could mitigate the growth of damaged cells during the tumor promotion period.

Cellular responses to DNA damage are context-dependent, with common outcomes ranging from repair to apoptosis [33]. Several cancer-resistant species respond to DMBA by initiating cell death via Caspase-3 cleavage, indicating activation of the intrinsic apoptosis pathway in damaged cells may protect against carcinogenesis [34,35]. We investigated whether DMBA application led to cleaved Caspase-3 expression in *Acomys*, which could provide an underlying reason for the lack of *Hras* mutations following a year of TPA. However, analysis of the skin 24 hours post-DMBA and seven days post-DMBA indicated that no epidermal apoptosis occurred due to DMBA-mediated DNA damage, although some cleaved Caspase-3 expression was noted in Integrin-alpha-6 (Itga6) negative dermal tissue adjacent to the epidermis in *Acomys* (Figure 2D-E).

Effects of TPA on DMBA-damaged keratinocytes

Given the similarities between γH2AX and cleaved Caspase-3 prevalence in *Acomys* and C57Bl/6J mice in response to the tumor initiation insult by DMBA, we hypothesized that an accumulation of damaged cells must occur before a tissue-wide response occurs. Thus, we assessed the skin during the tumor promotion stage of chemical carcinogenesis (Figure 3A). When it follows DMBA treatment, TPA is a potent tumor promoter that causes epidermal keratinocyte proliferation in *Mus*, shown by hyperplasia of the epidermis measuring 125µM after four topical applications (Figure 3B-C). This also occurs in *Acomys*, though to a lesser extent, with epidermal thickness increasing to 90µM with the same treatment of four TPA applications (Figure 3B-C).

The epidermis is composed of a basement membrane and basal layer that can be broadly defined by Integrin-alpha-6 (Itga6) expression and harbors the stem and progenitor cell populations of the interfollicular epidermis, and a differentiated suprabasal layer that can be identified through its expression of Keratin-10 (Krt10). We assessed the contribution of basal and suprabasal keratinocyte populations within the epidermis of *Acomys* and *Mus* to the overall hyperplasia. Typically confined to distinct regions with little mixing of layers, we found widespread co-localization of Itga6 and Krt10 in *Mus* (Extended Data Figure 2A), indicating a disorganized, pathogenic phenotype often found during aberrant proliferation [36]. In striking contrast, the tissue architecture and organization in DMBA/TPA-treated *Acomys* remained intact despite notable hyperplasia (Extended Data Figure 2A). These results indicate that *Acomys* maintains tight control over cell fate specification during periods of rapid proliferation.

Staining for cleaved Caspase-3 in hyperplastic DMBA/TPA-treated skin revealed extensive apoptosis in *Acomys*, but not in *Mus* (Figure 3C, D). To confirm that the antibody used was suitable for *Acomys*, specificity was demonstrated through western blot analysis of the same tissue, with cleavage events resulting in both 17kDa and 19kDa fragments as expected (Extended Data Figure 2B). These findings support the hypothesis that following an accumulation of damaged cells through DMBA and subsequent TPA applications, *Acomys* undergoes a tissue-wide response to remove the tumor-prone cells. However, the extent of apoptosis identified was unexpected, given the simultaneous finding of epidermal hyperplasia and intact epidermal tissue architecture (Figure 3B, Extended Data Figure 2A). Apoptosis has been closely linked to epimorphic tissue regeneration in several contexts, including *Xenopus* tail and planaria whole-body regeneration [37,38]. This suggests that the large regions of cleaved Caspase-3 positive cells may not only clear damaged cells from the epidermis but also elicit remodeling and regeneration of the existing epidermis[39].

If apoptosis was driving an epimorphic regenerative wound healing phenotype, we hypothesized that there would be an increase in epidermal proliferation concurrent with cleaved Caspase-3 expression. To determine this, we administered a pulse of 5-Ethynyl-2-Deoxyuridine (EdU) to the animals 2-hours prior to sacrifice. Indeed, *Acomys* epithelium showed a substantial increase in proliferation relative to *Mus* at the same timepoint (Figure 3B, E). Additionally, *Acomys* epithelial cells showed proliferation in both the basal and suprabasal layers, which was unusual compared to *Mus* (Figure 3B), in which proliferative cells are restricted to the basal layer in pathogenic and non-pathogenic situations. This suprabasal proliferative phenotype is reminiscent of embryonic epidermal development in *Mus*, where up to 60% of proliferative cells can be found in the suprabasal layers during epidermal stratification. In contrast, adult wound healing is driven by proliferation and migration of the basal layer and hair follicle stem cells into the wound bed [40,41]. The unique proliferation pattern identified in the epidermis of *Acomys* combined with high epidermal cleaved-Caspase3 and clear epidermal stratification strongly suggests tissue remodeling and replacement of cells damaged by DMBA (Figure 3F).

Erk-dependent regenerative wound healing phenotype in *Acomys*

One characteristic of scarless wound healing and epimorphic regeneration in *Acomys* is altered collagen organization and density compared to *Mus*, along with decreased tissue stiffness[13,29]. To interrogate whether this regeneration-supportive microenvironment was found in the DMBA/TPA-treated skin of *Acomys*, we employed Masson's Trichrome staining to identify collagen-rich regions in the tissue. A decrease in collagen density was found in the DMBA/TPA-treated *Acomys* dermis at the junction of the epidermal basement membrane relative to *Mus* (Extended Data Figure 2C), consistent with previous reports on *Acomys* skin and indicative of a tissue microenvironment that supports regeneration [6,42].

Recent findings have demonstrated that sustained Erk activity is responsible for generating a regenerative rather than a scarring response in *Acomys* skin. To determine if high p-Erk activity, consistent with a regeneration response, occurs in DMBA/TPA-treated *Acomys* skin, we used immunofluorescent staining with a validated p-Erk antibody [26]. We identified a large increase in p-Erk localized within the epidermis, which was absent in *Mus* with the same treatment and at the same timepoint (Figure 4B, 4C, Extended Data Figure 3A-B). Given the ability of Mek inhibition to successfully inhibit Erk activation, and thus wound healing and regeneration in *Acomys*, we hypothesized that a similar approach could inhibit the regenerative or remodeling processes activated in response to DMBA/TPA [26]. After validating that daily treatment with the MEK inhibitor trametinib successfully delays the closure of a 4mm punch wound to the ear (Extended Data Figure 3C-D), we tested whether trametinib treatment during DMBA/TPA applications would disable the remodeling response.

Acomys were administered trametinib daily while maintaining the DMBA/TPA treatment protocol (Figure 4A). Analysis of the skin tissue at the end of treatment confirmed that trametinib conferred a reduction in cutaneous p-Erk (Figure 4B-C). This was further associated with reduced epidermal hyperplasia compared to DMBA/TPA treatment alone (Figure 4D-E). We next investigated the effect of trametinib on cleaved Caspase-3 after DMBA/TPA treatment and found a reduction within the epidermis (Figure 4F-G). These findings confirm that hindering regeneration in *Acomys* prevents apoptosis of DMBA/TPA-damaged epithelial keratinocytes.

Effect of wound healing on tumor development

In laboratory mice, wound healing has previously been shown to accelerate DMBA/TPA-mediated tumor formation[43]. In *Acomys*, the data presented here suggest that the regenerative process that occurs in response to a wounding event may suppress tumor formation. Unlike the dorsal skin, the *Acomys* ear pinna regenerates through the formation of a blastema, and therefore may utilize alternative mechanisms during the healing process [9]. We tested the hypothesis that *Acomys* regeneration suppresses tumor formation in the ear pinna of *Acomys* by performing a 4mm punch biopsy one week after DMBA application and then implementing twice-weekly TPA treatments two days following the wound until wound closure (Figure 5A). DMBA/TPA treatments resulted in a slightly faster rate of tissue regeneration in the first two weeks following the biopsy punch, however this effect was short-lived, and the wounds ultimately healed at the same rate without any tumor formation (Figure 5B). Instead, mild epidermal hyperplasia was noted in both the regenerated ear pinna as well as the adjacent unwounded tissue, as expected after DMBA/TPA treatment (Figure 5C).

Cleaved Caspase-3 expression is a hallmark of carcinogen-mediated regeneration in *Acomys* dorsal skin, though it has not yet been described as a component of blastema-mediated regeneration, such as in ear pinna wounds. To investigate whether cleaved Caspase-3 expression can be found in healing ear tissue at similar rates to DMBA/TPA treated skin, we utilized our ear punch plus DMBA/TPA model with tissue collection 24 hours after the fourth TPA application, thus replicating conditions used in the dorsal skin (Figure 5D). Notably, we found an increase in the density of cleaved Caspase-3 expression at the wound edge in both epidermal and mesenchymal blastema tissue of the vehicle-treated control ear, thus confirming that apoptosis is an active component of the tissue regeneration process, even several weeks after the initial injury (Figure 5E). Similar to the findings in dorsal skin, four TPA applications to the wounded ear resulted in cleaved Caspase-3 expression readily found in epidermal keratinocytes and dermal fibroblasts (Figure 5C). These data provide further evidence that DMBA/TPA treatment induces a regenerative response in *Acomys*.

Single-cell mRNA sequencing

To further identify the dynamic cellular and molecular mechanisms of regeneration and carcinogenesis that occur during the early response to DMBA/TPA, single-cell RNA sequencing (scRNA-seq) was performed on untreated and DMBA/TPA-treated cells (Figure 3A, Extended Data Figure 4A) from *Acomys* and *Mus*. All major cell types expected within the skin were identified, including fibroblasts, keratinocytes from the hair follicle and interfollicular epidermis (IFE), and immune cells (Figure 6A-B). Fibroblasts were the most abundant cell type in both

Acomys and *Mus*, followed by T cells in *Acomys,* and macrophages in *Mus* (Figure 6A-B, Extended Data Figure 4B).

Recent work has highlighted the presence of a pro-regenerative fibroblast population in the regenerating ear pinna of *Acomys,* which is defined in part by its expression of Tenascin C (*TnC*)[26]. Given the abundance of fibroblasts within the skin, we identified an expression enrichment in *Acomys* of the previously identified regeneration-associated factors TnC, High Mobility Group AT-Hook 2 (*Hmga2*), Platelet Derived Growth Factor Alpha (*Pdgfrα*), Fibronectin (*Fn1*), *Twist1*, Midkine (*Mdk*), Paired Related Homeobox1 (*Prrx1*) and lectin galactoside-binding soluble1 (*Lgals*) (Figure 6C). Of these markers, TnC was particularly enriched in a subset of *Acomys* fibroblasts that were specific to DMBA/TPA treatment but nearly absent in *Mus* (cluster 7, Figure 6C, Extended Data Figure 4), indicating a treatment-dependent regeneration signature. This cluster is further defined by its expression of remodeling-associated genes, including the metalloprotease inhibitor *Timp1*, and a unique expression of Cochlin (*Coch*) (Figure 6C). This regeneration-associated fibroblast cluster is unique to DMBA/TPA-treated *Acomys*, with negligible TnC expression identified in *Mus* fibroblasts (Figure 6, Extended Data Figure 4B).

Tumor Necrosis Factor (*TNF)* signaling is a significant driver of cutaneous tumors, and mice deficient in TNF or its signaling receptor Tnfrsf1 show resistance to chemical carcinogenesis and tumorigenic clonal expansion [44–46]. Therefore, we hypothesized that *Acomys* likely expressed decreased *Tnf.* Interestingly, we identified greatly increased expression of *Tnf* in *Acomys*, primarily in T Cells. *Tnf* expression in *Mus* is found in T cells to a lesser extent, and is also found in neutrophils, mast cells, and macrophages, with macrophages being the primary contributor based on population size (Figure 6) [44]. The primary receptor for TNF, Tnfrsf1a, shows high levels of expression in treated epidermal cells in *Acomys* relative to *Mus* (Figure 6D, Extended Data Figure 4). Although these results are contrary to the assumed role of TNF signaling from a cancer perspective, they support a role for T cell-derived TNF as a driver of the MAPK-mediated regenerative processes and an increase in apoptosis, which mirrors processes observed in zebrafish [47–50].

Discussion

Here, we show that *Acomys* possesses the intrinsic ability to resist cutaneous tumor development, explicitly due to its regenerative and tissue remodeling properties. This is the first study, to our knowledge, to show that tissue regeneration in mammals can confer cancer resistance. We demonstrate that the tissue-wide response of *Acomys* dorsal skin to chemical carcinogenesis is indicative of epimorphic regeneration, which ultimately leads to the elimination of cells with tumorigenic potential through the widespread apoptosis of damaged keratinocytes. This phenotype is reminiscent of tissue remodeling during injury-induced regeneration in Xenopus and planaria [37,51] and is in stark contrast to that of the non-regenerative laboratory mouse *Mus* *musculus*, which readily develops cutaneous tumors without apoptotic cells present in the early stages of carcinogenesis. *Acomys'* ability to resist tumor formation largely depends on p-Erk signaling, a critical mediator of epimorphic regeneration and stem cell survival following DNA damage [26,52].

It will be of great interest to determine whether a similar response to carcinogen or DNA damage is seen in other *Acomys* tissue types, as well as further investigate the cellular and molecular processes that result in altered immune cell recruitment and activity as identified in our scRNA-seq analysis. Nevertheless, this study, along with ongoing work by many groups to better define the mechanisms of epimorphic regeneration in spiny mice, opens the door to further define and translate data found in this remarkable species to human patients. Cancer is the second leading cause of death in the US and the world, and patients are in need of the completely novel methods of prevention and intervention that *Acomys* may reveal.

Materials and Methods

Animals

Spiny mice (*Acomys* spp.) were bred from the colony of animals maintained at Cornell University. C57Bl/6J and FVB/N (*Mus musculus*) were originally purchased from the Jackson Labs (#000664, #001800) and maintained in the White Lab breeding colony. All animals were housed with littermates of the same sex, up to a maximum of three (*Acomys*) or five (*Mus*) animals per cage, at Cornell University, Ithaca, NY. Food and water were provided ad libitum throughout the study. Spiny mice were fed a 14% protein diet (Teklad), and C57Bl/6J and FVB/N animals were fed standard mouse chow (Teklad). All animals were kept on a 14:10 light:dark cycle. Only female mice were used for tumor initiation and progression studies (Figure 1). For all remaining data, both male and female animals were used with no evident sex effect.

Upon experimental endpoints, animals were injected with a single dose of 0.33mg EdU (Cayman Chemical #20518) by intraperitoneal (I.P) injection 2 hours prior to euthanasia by carbon dioxide overdose.

All animal treatments and procedures were approved by the Cornell University Institutional Animal Care and Use Committee (IACUC) under protocol #2014-0096.

DMBA/TPA

All topical treatments began in adult animals during the telogen phase of the hair cycle (*Acomys* 9-14 weeks, *Mus* 8-10 weeks). Animals were anesthetized with vaporized isoflurane, and their dorsal skin was shaven with electric pet hair trimmers (Wahl #9868).

DMBA (Cayman Chemical #30383) was dissolved in acetone to a final concentration of 400nmol and was applied evenly to the shaved region by pipetting. One week after DMBA application, TPA (Cayman Chemical #10008014) was diluted to a final concentration of 40nmol in ethanol and topically applied evenly to the treatment region by pipetting. TPA treatments continued until the animals reached a designated endpoint (either a specific number of treatments, tumor diameter > 4mm or one year).

Ear Punch

Animals were anesthetized by vaporized isoflurane inhalation. A 4mm biopsy punch was used to excise a region from the proximal ear, approximately 2.5mm from the skull. The wound region was measured by caliper daily during the healing process along the x and y diameter axes. To calculate the wound area, the radius (r) was calculated from measured diameters and used in the formula: $A=\pi(r_x r_y)$.

Trametinib

The Mek inhibitor Trametinib (MedChem Express #HY-10999) was diluted in 10% DMSO, 90% sterile saline and administered to animals at 1.4mg/kg by I.P. injection. Trametinib was administered daily during ear wound healing and during short-term DMBA/TPA treatments (3 weeks).

Histology and Immunohistochemistry

During tissue collection, fresh skin was embedded in optimal cutting temperature compound (OCT, Tissue-Tek #4583). Eight (8)-µM sections were cut and fixed for 10 minutes with 10% neutral buffered formalin prior to antibody or hematoxylin and eosin staining.

Formalin-fixed paraffin-embedded (FFPE) tissues were fixed overnight in 10% neutral buffered formalin at 4°C, washed, dehydrated by increasing ethanol dilutions, and cleared in xylene-substitute (VWR #89370-090) prior to overnight incubation in liquid paraffin. Five (5)-micron sections were cut and rehydrated. Masson's trichrome staining (Polysciences #25088-100) was performed according to the manufacturer's instructions.

For all antibody stains, normal donkey serum (Equitech Bio #SD30-0500) was diluted to 10% in PBS plus 0.1% Tween-20 for blocking. Slides were mounted using Fluoroshield Mounting Medium with Dapi (Abcam #ab104139).

Hematoxylin and eosin staining was performed according to the manufacturer's instructions using the Hematoxylin and Eosin stain kit (Vector Laboratories #H-3502).

Click-It EdU Plus (Invitrogen #C10637) was used according to the manufacturer's instructions following antibody staining.

Brightfield and fluorescence microscopy was performed on a Leica DM2500 upright microscope and DFC7000T camera. All image analysis was performed in ImageJ (RRID: SCR_002285).

Antibody	**Dilution (IF)**	**Dilution (WB)**	**Vendor/product number**
Cleaved Caspase-3 (Rabbit, monoclonal)	1:400	1:1000	Cell Signaling Technologies #9664
p-Erk 1/2 (Rabbit, monoclonal)	1:200	1:1000	Cell Signaling Technologies, #4370
Gapdh (Rabbit, monoclonal)		1:1000	Cell Signaling Technologies, #2118
yH2AX (Rabbit, monoclonal)	1:800		Cell Signaling Technologies, #9718
Integrin Alpha-6/CD49F (Rat, monoclonal)	1:200		Biolegend, #313602
Keratin-6 (Rabbit, polyclonal)	1:800		Biolegend, #905701
Keratin-10 (Rabbit, polyclonal)	1:100		Biolegend #905404
Keratin-14 (Rabbit, polyclonal)	1:100		Biolegend, #905304/905303

FFPE DNA Isolation and Sequencing

FFPE tissue blocks were sectioned at 8μM. Six sections were placed immediately into a 1.5ml Eppendorf tube containing 200ul of xylenes (Thermo #422685000). DNA was extracted with the E.Z.N.A FFPE DNA Kit (Omega Biosciences #D3399-01) according to manufacturer instructions. Primers for *Acomys Hras* amplification were designed in SnapGene (RRID: SCR_015052) software against the NCBI reference genome for *Acomys cahirinus*. The PCR product was gel-purified (Omega Biosciences #D2500-01) and submitted to the Cornell Biotechnology Resource Center (BRC) for Sanger Sequencing.

Protein Isolation and Western Blot

Freshly collected samples were snap-frozen in liquid nitrogen and stored at -80°C prior to processing. Lysis and sonication were performed in 1x cell lysis buffer (Cell Signaling Technologies #9803) with a final concentration of 1mm PMSF (Thermo Scientific #36978) and Halt protease and phosphatase inhibitor cocktail (Life Technologies #78441).

Protein was quantified using Pierce BCA assay (Thermo Scientific #23225), reduced at 100°C for 10 minutes in 1x Lamelli buffer (BioRad #1610747) with 2-mercaptoethanol (Sigma #M6250). Gels were hand-cast using TGX FastCast Acrylamide Kit (12%, BioRad). Equal amounts of protein were run (20ug) and transferred onto polyvinyl difluoride (PVDF) membranes. Membranes were blocked in 5% bovine serum albumen and incubated with primary antibody diluted at 1:1000 (unless otherwise noted in the table) in blocking solution at 4°C overnight. Horseradish peroxidase-conjugated secondary antibodies were diluted 1:3000 in blocking solution to probe primary antibodies and were detected with Clarity ECL substrate (BioRad #1705061).

Western blot quantification was performed in ImageJ with the analyze gel plugin.

Statistical Analysis

Statistics were performed in GraphPad Prism (RRID: SCR_002798). Graphs are displayed as means +/- SEM or means +/- SD, as the corresponding figure legend indicates. The number of animals used for each analysis is annotated in the figure legend as n. The two-tailed Student's T-test was used when two groups were compared, and the 2-Way ANOVA test was performed when three or more groups were compared. Statistical significance is defined as a p-value of less than 0.05, and the degree of significance is indicated in the figure legends.

Single-cell RNA sequencing

Dorsal skin tissue was harvested from age- and sex-matched *Mus* and *Acomys* that were untreated or 24 hours post-DMBA plus four (4) TPA treatments. Three (3) biological replicates per species per timepoint were collected, and cells were pooled for library preparation. To prepare the cell suspension, isolated skin tissue was digested in 20mg/ml collagenase type I (LS004197, Worthington) in serum-free keratinocyte cell media and incubated at 37 °C for one and a half hours with gentle shaking with 0.05mg/ml DNase I (Alfa Aesear #J62229) added in the last half hour. The tissue suspension was triturated and passed through a 70um cell strainer, and red blood cells were lysed with ACK lysis buffer (Gibco #A1049201). Due to the large amount of apoptosis in DMBA/TPA-treated *Acomys* skin, all samples were cleaned with a dead cell removal MACS kit (Miltenyi Biotec #130-090-101). After cleanup, cell viability was assessed by trypan blue staining at 95% on average.

Single-cell suspensions were submitted to the Cornell University BRC Genomics Core Facility (RRID:SCR_021727), where they were loaded into the Chromium X instrument (10x Genomics) for single-cell barcoding and subsequent library preparation using Chromium Single-Cell 3' reagents. Libraries were sequenced on a NovaSeqX-25B instrument with an average of 75k reads per cell. An *Acomys* reference was generated using Cell Ranger (v8.0.1) mkref pipeline and genome assembly mAcoDim1_REL_1905 available through Ensembl. Fastqs were processed with Cell Ranger count to either the mm10 genome indexes or the generated *Acomys* index. Downstream processing, including quality control and species-specific UMAP clustering, was performed in Seurat (v5.0.2). Cells with a nFeature_RNA greater than one and less than 9000 were included for clustering. In *Mus*, cells with a mitochondrial content greater than 20% were excluded. No mitochondrial gene filter was applied to *Acomys* samples due to a lack of annotated mitochondrial genome availability. Clustering was performed based on the first 50 principal components (PCs), and sample integration was performed using Harmony (v1.2.0) and reclustered (*49*). Cell clusters were identified based on their treatment-independent conserved gene expression, as well as treatment-induced gene expression.

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Project Administration: LRD, ACW
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Writing – original draft: LRD, ACW
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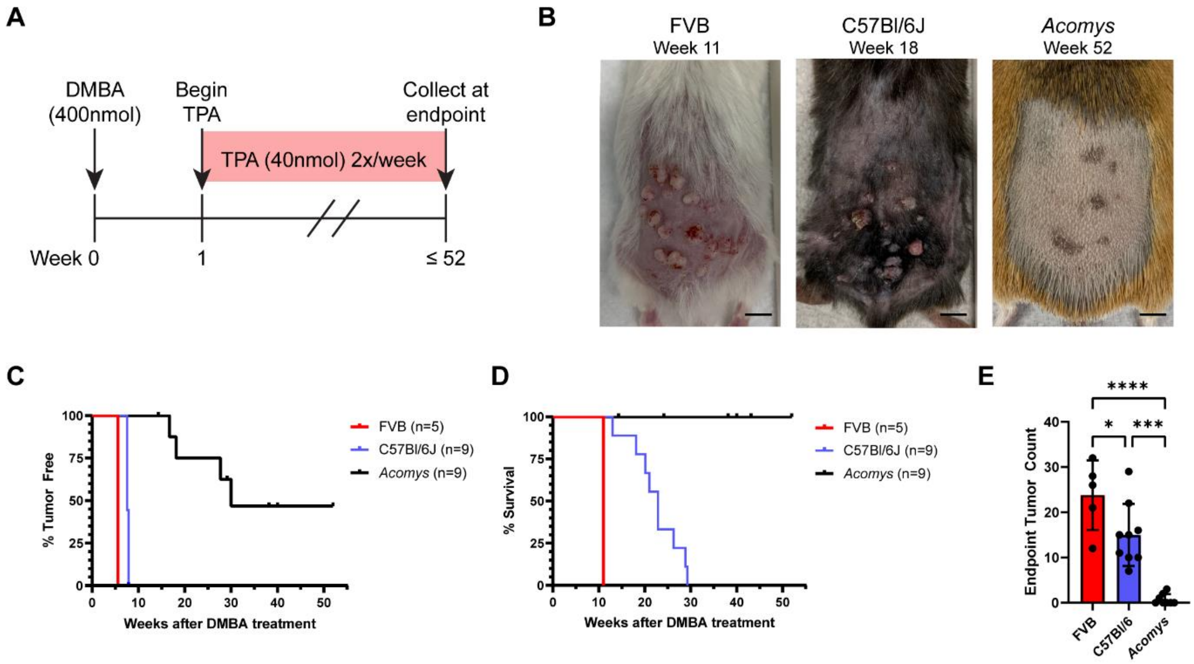


Figure 1 – Tumor onset and progression are delayed, and tumor severity is reduced in DMBA/TPA-treated *Acomys*. **(A)** DMBA/TPA treatment timeline **(B)** Mouse images at endpoint. Scale bar equal to 1cm. **(C)** Tumor-free survival curves showing time until visible neoplasia. **(D).** Survival to clinical endpoint (defined as tumor larger than 4mm, most tumors ulcerated, or at one year of treatment – whichever comes first). **(E)** Number of tumors per animal at clinical endpoint. Each point represents an independent biological replicate and error bars indicate standard deviation. Ordinary one-way ANOVA used to determine statistical significance. *=p<0.05, **=p<0.01, ***=p<0.001. FVB n=5, C57Bl6/J n=9, *Acomys* n=9.

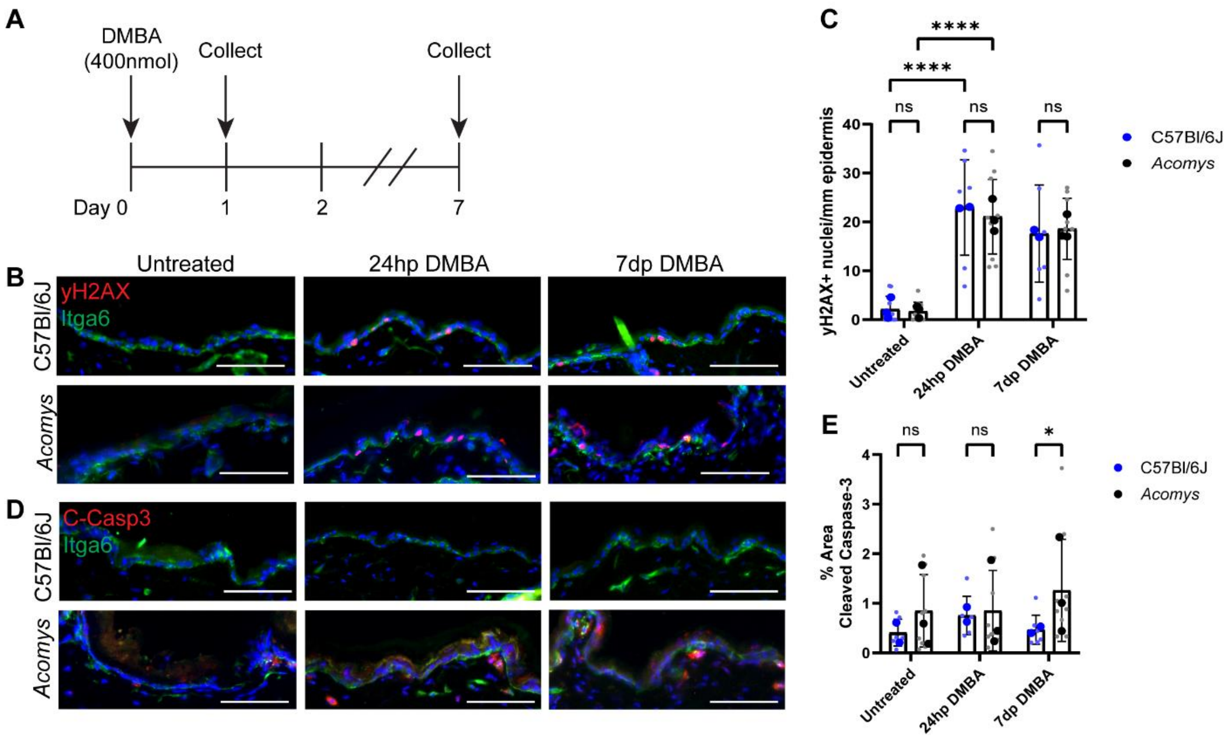


Figure 2 – Similar immediate response to DMBA in *Acomys* and *Mus.*

(A) C57Bl/6J and *Acomys* are treated with 400nmol of DMBA and collected either 24 hours post (24hp) or seven days post (7dp) application **(B)** Epidermal DNA damage at 24-hours post-DMBA and 7-days post-DMBA in C57Bl/6J and *Acomys* skin. The epidermis is marked by integrin-alpha-6 (Itga6) and DNA damage by yH2AX, blue nuclear counterstain with DAPI**.** Color channels in fluorescence images uniformly brightened in Photoshop. **(C)** Number of yH2AX positive keratinocytes per mm of epidermis **(D)** Apoptosis marked by cleaved Caspase-3 (C-Casp3) expression. The epidermis is marked by Itga6 and blue counterstain with DAPI. Cleaved Caspase-3 positive cells can be found underneath the epidermis following DMBA treatment, but not within. Color channels in fluorescence images uniformly brightened in Photoshop. **(E)** Percent area of cleaved Caspase-3 throughout the skin tissue. Data in Figure 2C and 2E are presented as the mean value of each animal assessed (dark, large dot), quantified from five fields of view per animal (small light dot). Error bars indicate the standard deviation. Statistics measured by 2-way ANOVA, ****=p<0.0001. *=p<0.05. C57Bl/6J n=3 (untreated), n=2 (24hp and 7dp DMBA). *Acomys* n=3 for all timepoints.

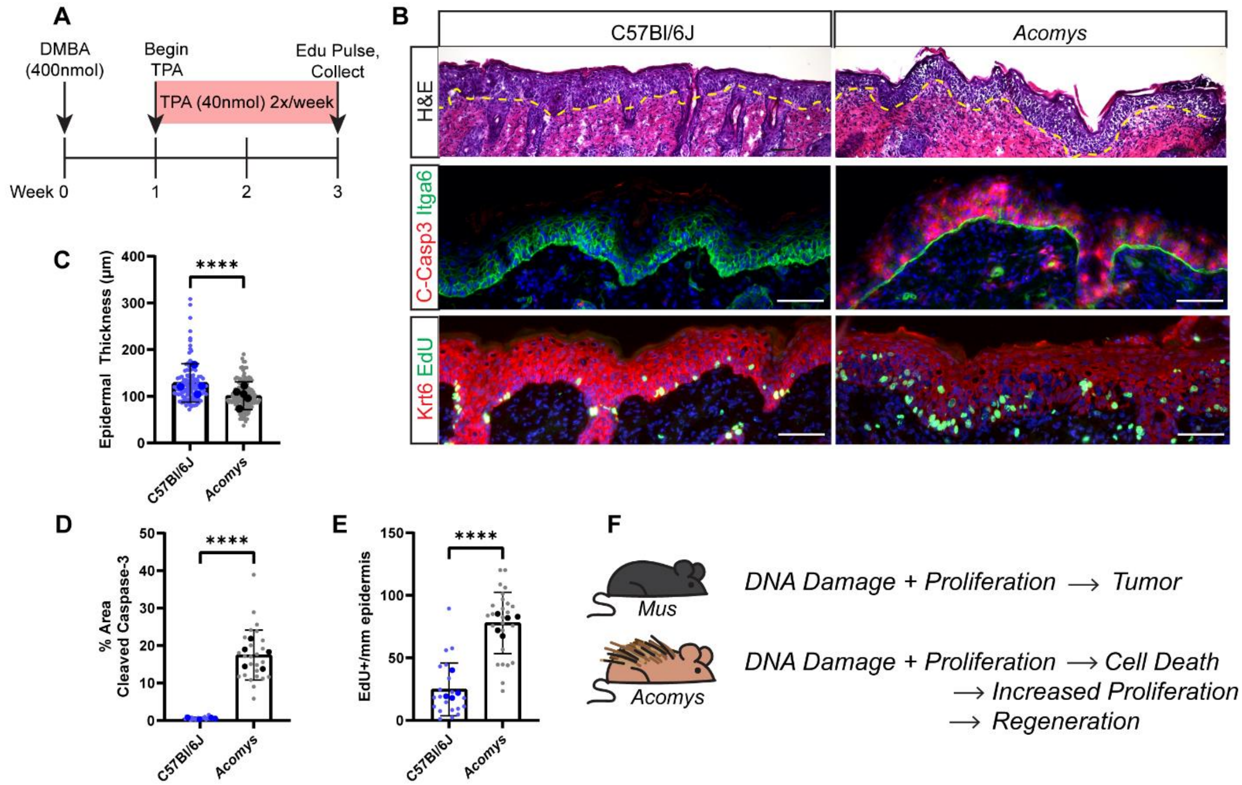


Figure 3 – DMBA/TPA induces a regeneration-like phenotype in *Acomys.*

(A) C57Bl/6J (n=4) and *Acomys* (n=5) are treated with 400nmol of DMBA, then after a week rest period, are treated with 2 TPA treatments per week for a total of 4 TPA applications. A single pulse of EdU is administered 2 hours before collection. **(B)** Histology analysis of DMBA/TPA treated *Mus* (C57Bl/6J) and *Acomys.* Hematoxylin and Eosin (H&E) staining reveals that epidermal thickness increased at similar rates between species. The yellow dashed line marks the epidermal–dermal junction. DMBA/TPA induces epidermal apoptosis in *Acomys,* but not in C57Bl/6J mice, as marked by cleaved Caspase-3 expression. EdU incorporation indicates a greater incidence of keratinocyte proliferation in *Acomys.* Color channels in fluorescence images uniformly brightened in Photoshop. **(C)** Epidermal thickness measurements are derived from 3 fields of view per animal (C57Bl/6J n=4, *Acomys* n=5), ten measurements per field of view. Each measurement is presented as a small, light-colored dot, and the average measurement per animal is presented as a large, dark dot. **(D)** DMBA/TPA induces epidermal apoptosis in *Acomys,* but not in C57Bl/6J mice, as marked by cleaved Caspase-3 expression. **(E)** EdU incorporation indicates a greater incidence of keratinocyte proliferation in *Acomys.* For **(C-E)**, measurements are derived from five fields of view per animal (C57Bl/6J n=4, *Acomys* n=5). Each measurement is presented as a small, light dot, and the average expression per animal is indicated as a large, dark dot. Error bars represent the standard deviations. Statistics measured by Student's T-Test, ****=p<0.0001. **(F)** Summary schematic of species-specific response to DMBA/TPA.

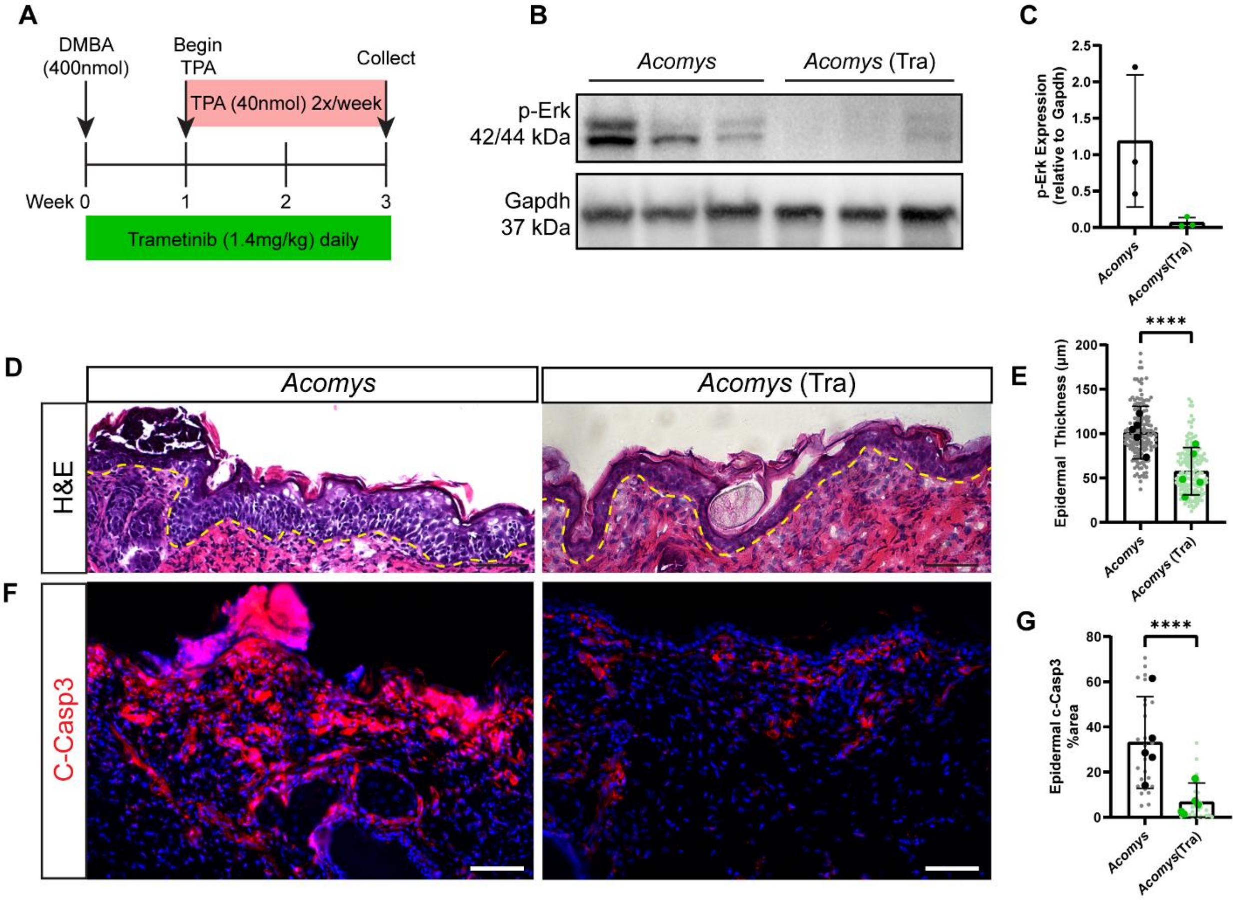


Figure 4 – Cutaneous hyperplasia and apoptosis following DMBA/TPA are reduced with Mek inhibition.

(A) A single dose of DMBA is applied topically to the dorsal skin of *Acomys* (n=3). One week after DMBA, TPA is applied topically twice per week for a total of four applications. Daily trametinib treatment begins with the first DMBA application and continues until the experimental endpoint. **(B)** Western blot of p-Erk and Gapdh in *Acomys* skin with DMBA/4xTPA treatment (left three lanes) and *Acomys* skin with DMBA/4xTPA+trametinib (Tra) (right three lanes). Blots are from two different gels with equal amounts of protein loaded and ran simultaneously. **(C)** Though p-Erk expression is variable in DMBA/4xTPA treated *Acomys*, there is a nearly complete absence of p-Erk in the skin of animals treated with trametinib. **(D-E)** Hematoxylin and eosin stain of DMBA/4xTPA and DMBA/4xTPA+trametinib (Tra) treated *Acomys* shows a moderate reduction in epidermal hyperplasia. The epidermal-dermal junction is marked with a yellow dashed line. Measurements are derived from three fields of view per animal (*Acomys* n=5, *Acomys* (Tra) n=5), ten measurements per field of view. Each measurement is presented as a small, light-colored dot, and the average measurement per animal is presented as a large, dark dot. **(F-G)** Cleaved Caspase-3 staining in DMBA/4xTPA and DMBA/4xTPA+trametinib (Tra) *Acomys*. Trametinib treatment causes a reduction in epidermal apoptosis as measured by cleaved Caspase-3 (*Acomys* n=5, *Acomys* (Tra) n=5). Color channels in fluorescence images uniformly brightened in Photoshop. Quantifications are performed on five fields of view per animal and are plotted as the small, light-colored dots on the bar graph. Error bars represent standard deviations. The average cleaved Caspase-3 expression per animal is plotted as a large, dark-colored dot on the bar graph. Student's T-Test is used to determine significance where $p<0.0001$ = ****. *Acomys* datapoints in panel E are also represented in Figure 3.

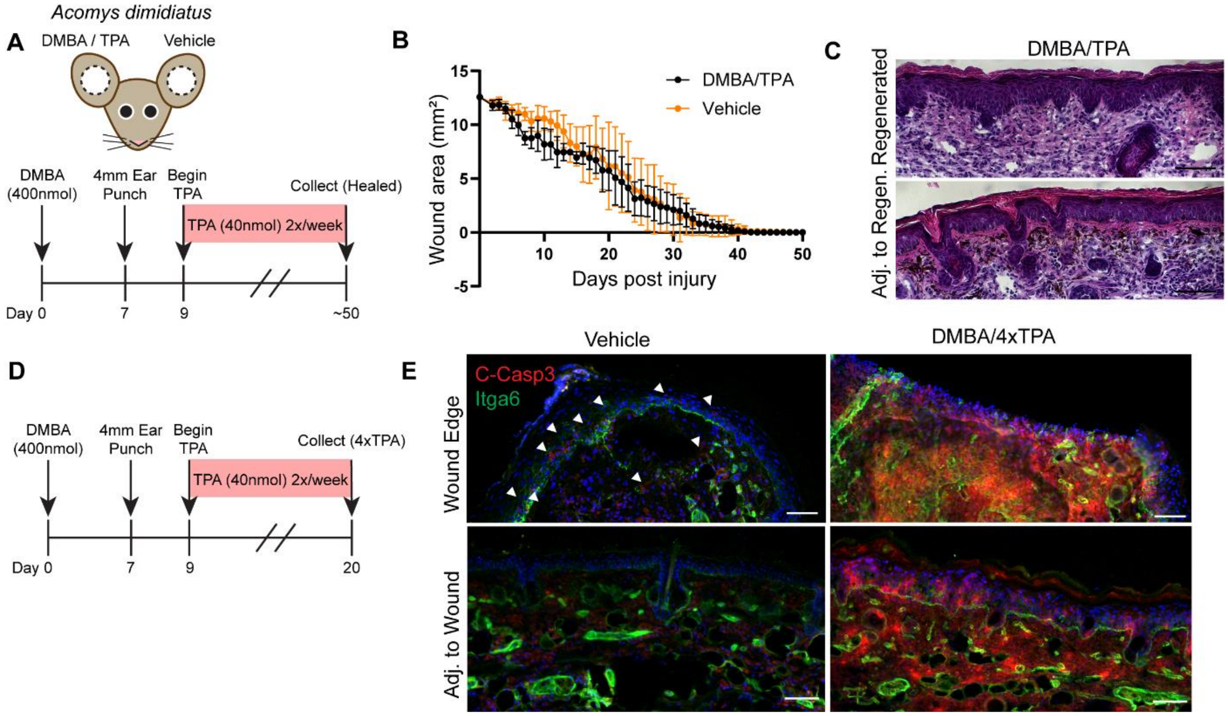


Figure 5 – Tumor formation in *Acomys* is not accelerated by excisional wound healing, nor is regeneration hindered by DMBA/TPA.

(A) To assess the contribution of DMBA/TPA to ear pinna regeneration, DMBA or vehicle control is applied to the left and right ears of *Acomys*, respectively. One week following DMBA/vehicle, a 4mm punch biopsy is taken from each ear. Two days after punch biopsy, a twice-weekly regimen of TPA or vehicle control is applied to the left and right ear, respectively, until healed (~50 days of treatment). **(B)** The wound area was measured daily until both the DMBA/TPA-treated and vehicle control-treated punch wounds had completely healed. Data are presented as the mean wound area and standard error of the mean. n=6 animals per group through day 13 post-injury, n=3 animals per group from day 13-endpoint. **(C)** Hematoxylin and eosin staining of regenerated ear tissue 24 hours after TPA treatment. Tissue was collected within 48 hours of both ear wounds completely shutting. The regenerated tissue is largely depigmented, but hair follicles and chondrocytes can be found throughout. Neither the regenerated tissue nor the adjacent unwounded tissue shows any signs of tumor development (n=3). **(D)** To assess the prevalence of cleaved Caspase-3 during blastema-driven wound healing and DMBA/TPA application in the ear pinna, DMBA or vehicle control is applied to the left and right ears of *Acomys*, respectively. One week following DMBA/vehicle, a 4mm punch biopsy is taken from each ear. Two days after punch biopsy, a twice-weekly regimen of TPA or vehicle control is applied to the left and right ear, respectively, and collected 24 hours following the final TPA application. **(E)** Tissue is collected following four TPA treatments and assessed for cleaved Caspase-3 at the wound edge (top panel) and at an uninjured site adjacent to the wound region. In the vehicle control-treated wound edge, cleaved Caspase-3 is found in keratinocytes at the leading edge (white arrowheads). With DMBA/TPA, cleaved Caspase-3 is widely expressed throughout the entire treated area, regardless of proximity to the wound edge. n=3 animals per group. Color channels in fluorescence images uniformly brightened in Photoshop.

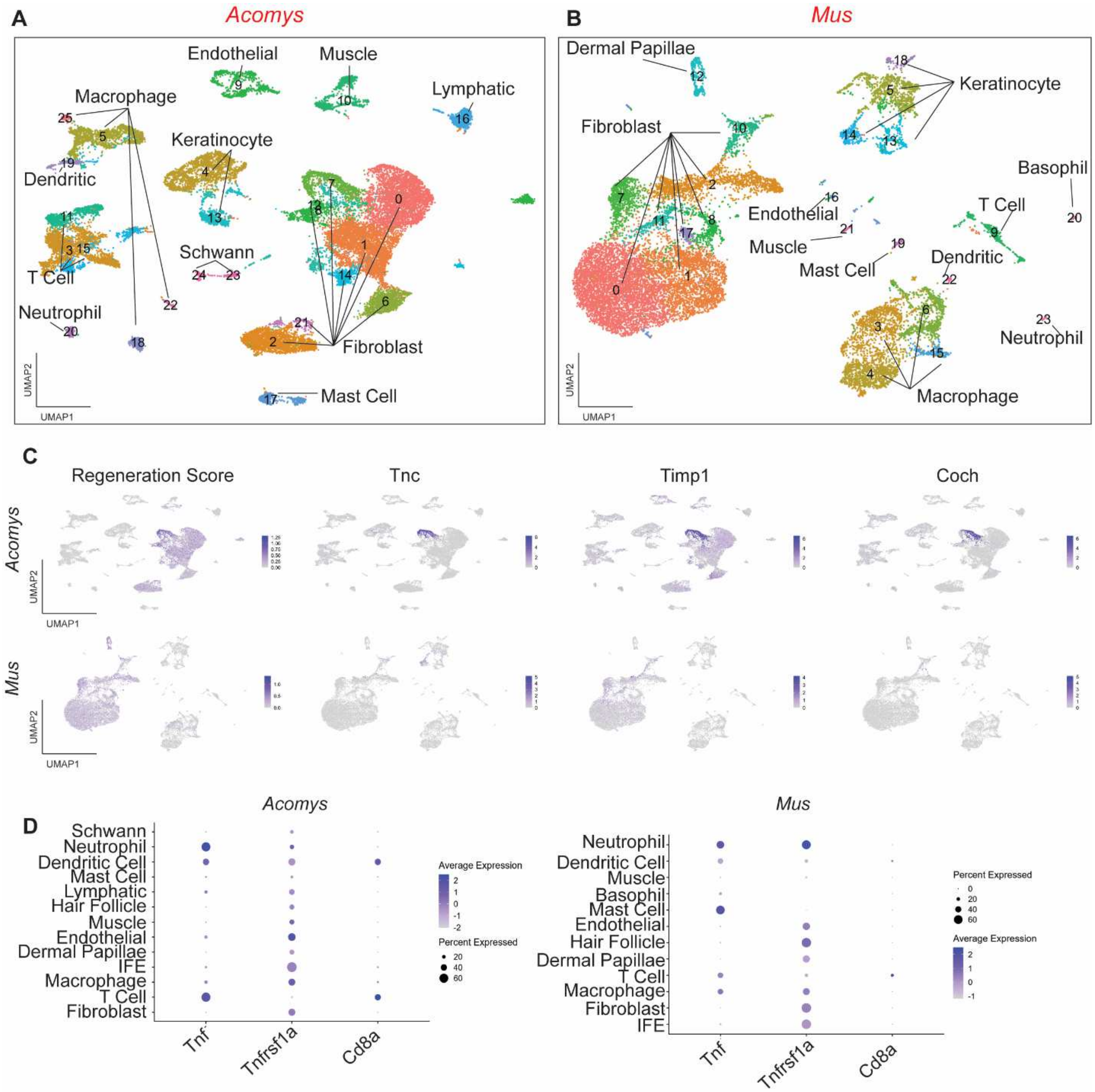
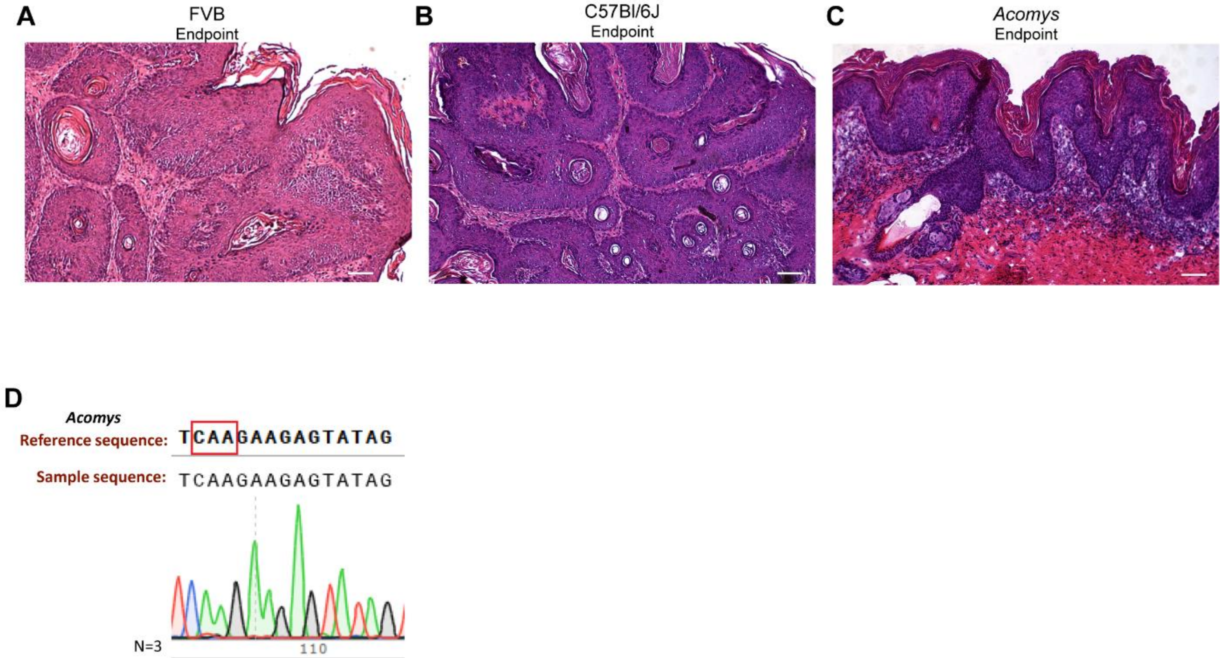
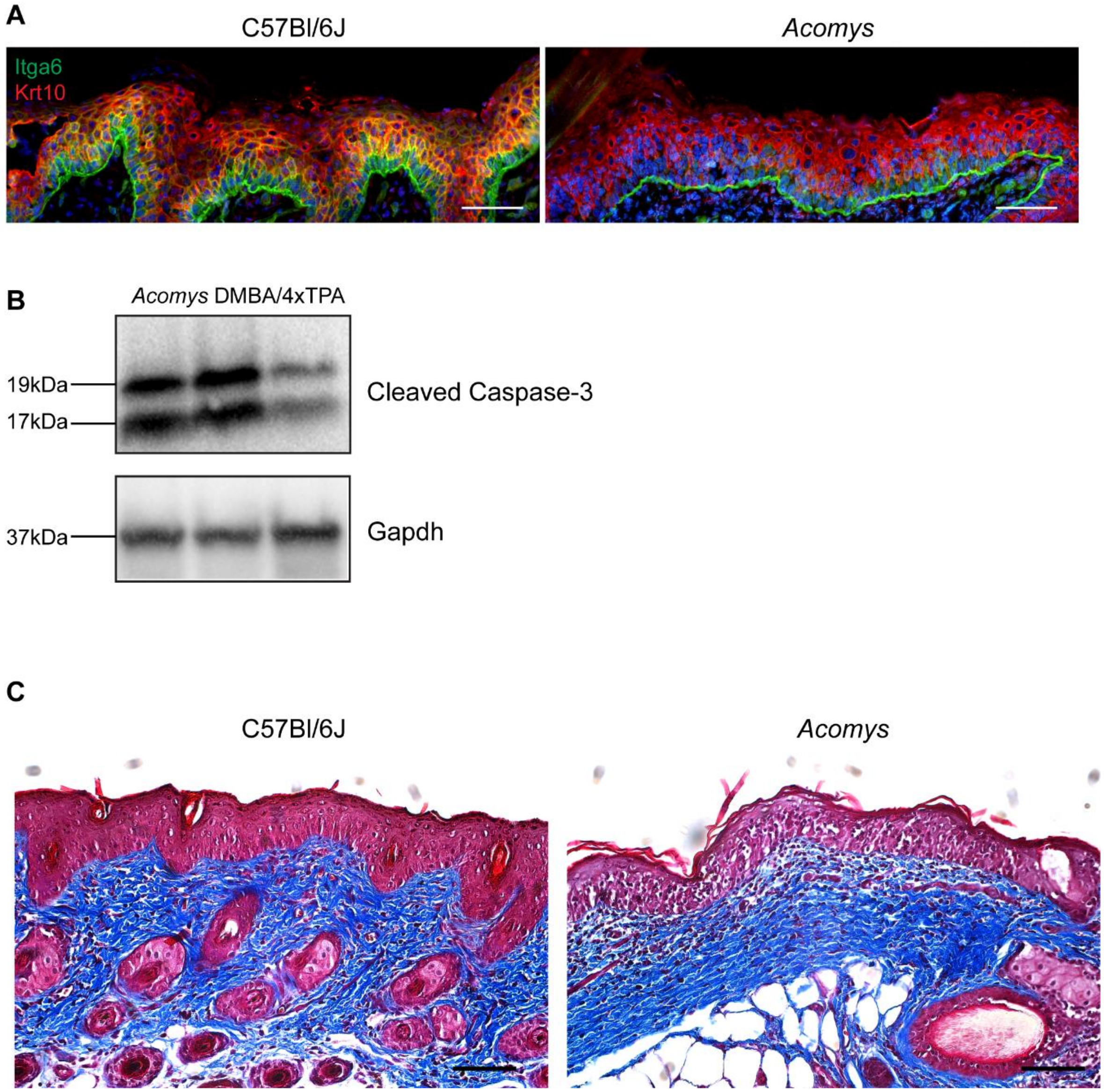


Figure 6 – Single-cell analysis reveals regenerative fibroblasts and altered T Cell signaling.

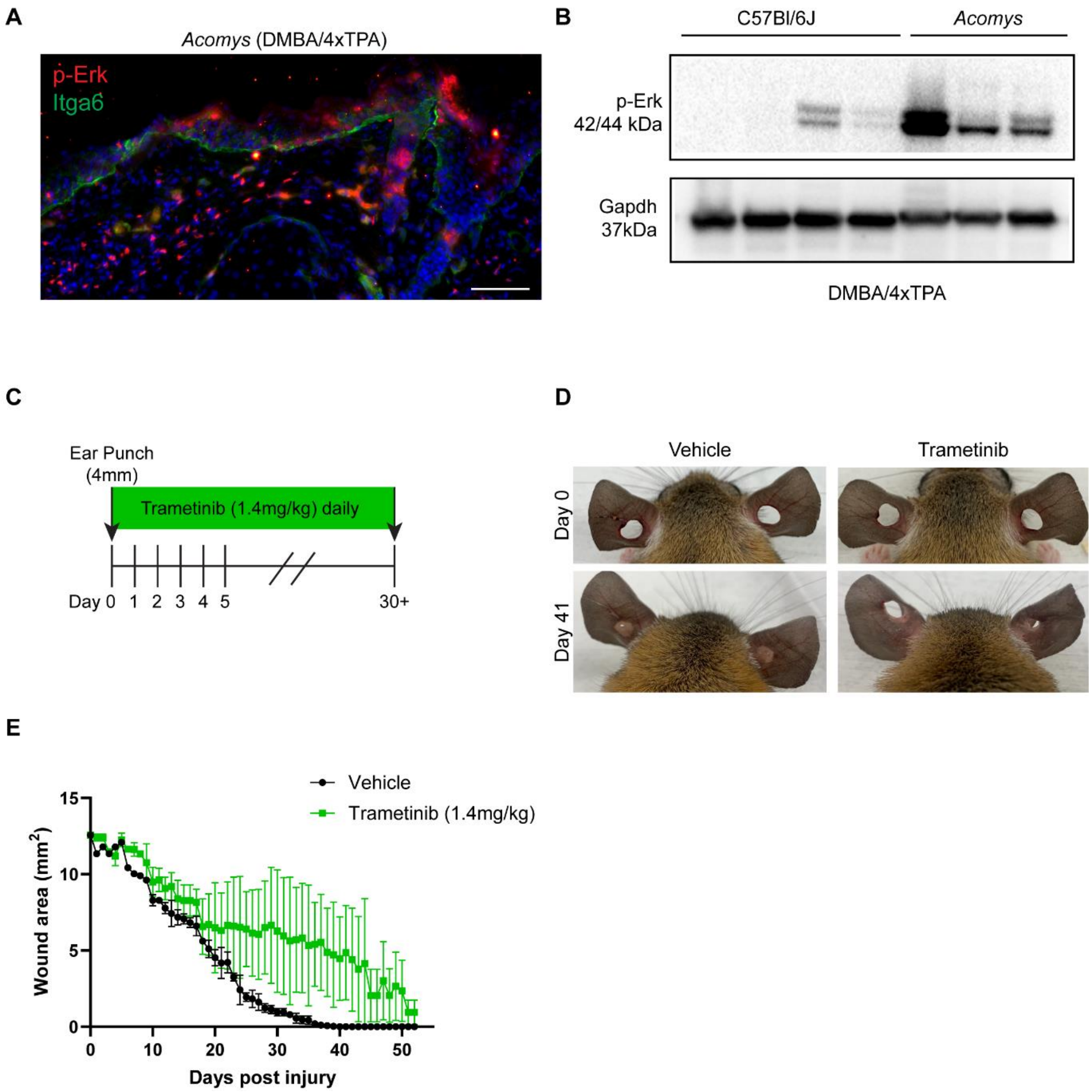
(A) UMAP projection of *Acomys* and **(B)** *Mus* cells derived from both untreated and DMBA/TPA-treated dorsal skin (n=3 per condition). **(C)** Feature plots of markers associated with regenerative fibroblast markers. Regeneration score is the composite expression of *Tnc, Hmga2, Pdgfrα, Fn1, Twist1, Mdk, Prrx1, and Lgals*. In *Acomys*, regeneration-specific marker *Tnc* is found primarily in DMBA/TPA-treated fibroblast Cluster 7, along with *Timp1* and *Coch*. **(D)** *Tnf* expression is produced primarily by the T cell population in *Acomys*. *Tnfr1a* is expressed at slightly higher levels in interfollicular epidermal (IFE) cells in *Acomys* relative to *Mus*. A larger population of CD8+ T cells is also found in DMBA/TPA-treated *Acomys*.



Extended Data Figure 1 – Tissue analysis at DMBA/TPA treatment endpoint. Representative tumor histology from **(A)** FVB/N (n=5 animals). **(B)** C57Bl/6J (n=5 animals) **(C)** *Acomys* (n=9 animals). Scale bars are equal to 100um. **(D)** Sanger sequencing of DMBA/TPA-treated tissue from *Acomys* (n=3) revealed no *Hras* mutation at amino acid 61. The red square in the reference sequence indicates the coding region of amino acid 61.

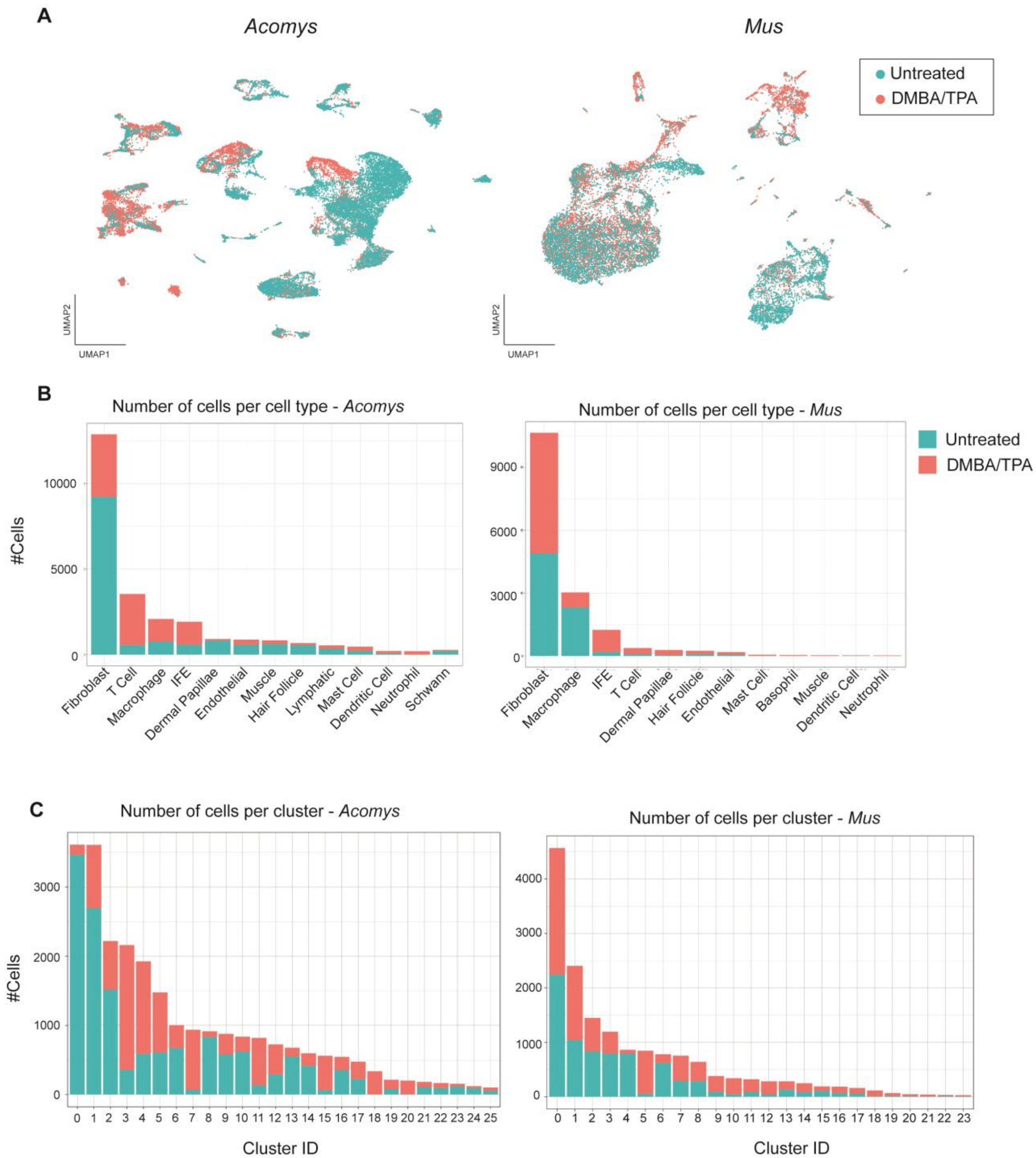


Extended Data Figure 2 - Tissue analysis with short-term DMBA/TPA treatment. For all data shown in this figure, the DMBA/TPA treatment scheme described in Figure 3A was applied (single DMBA treatment and four TPA treatments over two weeks). **(A)** In C57Bl/6J (n=4), basal cell marker Itga6 and differentiated cell marker Krt10 are co-expressed in the hyperplastic epidermis. In *Acomys* (n=3), Itga6 and Krt10 are expressed in their respective populations, but no co-localization is found, as indicated by the lack of yellow double-labeled cells. Color channels in fluorescence images uniformly brightened in Photoshop. **(B)** Western blot analysis of cleaved Caspase-3 in *Acomys* skin with DMBA/4xTPA shows antibody specificity and both the 17 and 19 kDa fragments of activated Caspase-3. Each lane represents a unique biological replicate. **(C)** Masson's Trichrome staining of C57Bl/6J (n=4) and *Acomys* (n=3). Collagen, identified by blue staining, is less dense at the junction of the epidermis and dermis in *Acomys* compared to *Mus*. Scale bars are equal to 100um



Extended Data Figure 3 – DMBA/TPA-mediated p-Erk expression in *Acomys,* but not *Mus,* indicates a regenerative response.

(A) Immunostaining for p-Erk (red) and Itga6 (green) in *Acomys* (n=3) skin shows p-Erk in both the epidermis and dermis following DMBA/4xTPA treatment. Scale bar equal to 100um. Color channels in fluorescence images uniformly brightened in Photoshop. **(B)** Western blot of C57Bl/6J (n=4) and *Acomys* (n=3) skin following DMBA/4xTPA treatment shows an increase in p-Erk in *Acomys* relative to C57Bl/6J. Each lane represents a unique biological sample. **(C)** To validate whether p-Erk inhibition via trametinib can suppress tissue regeneration in *Acomys*, a 4mm punch biopsy was performed on both the right and left ears. Trametinib or the vehicle control was administered daily until the tissue was regenerated. **(D)** Ear pinna regeneration is evident on day 41 in the vehicle control animal, but not the trametinib-treated animal. **(E)** Wound area over time. Points are displayed as the mean area and standard deviation. The left and right ear of a single animal is measured for each treatment group.



Extended Data Figure 4 – scRNA-seq profiles.
(A) UMAP projections of *Acomys* (left) and *Mus* (right), split by treatment (untreated or DMBA/TPA-treated). **(B)** Number of cells identified within the scRNA-seq dataset per cell type in *Acomys* and *Mus* split by treatment. **(C)** Number of cells identified within the scRNA-seq dataset by cluster ID (Figure 6) and split by treatment.

Supplementary Files

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